

is suitable for providing power for silicon based chips which have a high current requirement, such as GPUs, CPUs, IPU, high-end FPGAs, and high-performing ASICs often found in the latest AI applications.

Flex Power Module

<http://www.flexpowermodules.com>

Exposed power module

EMIPAK PressFit power module from Vishay Semiconductors is designed to facilitate a reliable performance in rugged 15 to 150A applications.



The module is optimised to offer higher performance at low tem-

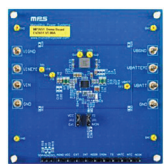
peratures. The module also features a temperature sensor for monitoring the temperature. It has an exposed substrate and optimised layout on the package that help to minimise stray parameters, allowing improved performance. The exposed design of the device also makes it easier for heat to vent out, which further improves the performance as well the product life.

Vishay Semiconductors

<https://www.vishay.com>

Evaluation board for charger IC

EV2651-VT-00A is an evaluation board for the MP2651 buck-boost charger IC. The MP2651 IC is designed for battery packs with one cell



to four cells in series. The device can accept an input voltage (VIN) of 4 to 22V to charge the battery. It also supplies a

wide 3 to 21V voltage range at the IN pin in source mode. This function is compliant with USB PD specifications. With the I2C/SMBus interface, the MP2651 can flexibly configure the charge and discharge parameters. The module can provide a charge current of up to 6A. It works at a switching frequency of 600kHz and offers an

efficiency of up to 93.09%.

Monolithic Power Systems

<https://www.monolithicpower.com>

Power converters for transport

The ERM100 and ERM150 series are compact quarter-brick, fully encapsulated 100W and 150W DC/DC modules optimised for rail and mass



transportation applications.

The ERM100 and ERM150 quarter brick

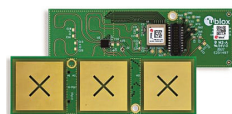
modules comply with multiple industry standards and provide reliable, high-efficiency solutions for rail system designers. Both converters have a European Railway Standard EN50155 compliant input voltage range of 36 to 160V DC. The modules have an efficiency of up to 91.5%, operate with a fixed switching frequency, and deliver a single, continuous power output. The ERM100 and ERM150 have no minimum load requirement and exhibit low ripple and noise characteristics. A baseplate that is optimised for contact cooling simplifies thermal management, while a heatsink version is available for designs requiring more stringent heat management.

Advanced Energy

<https://www.advancedenergy.com>

Asset tracking module

u-Blox ANT-B11 is the smallest antenna board which can accurately track an object using Bluetooth Angle-of-Arrival technology. The angle calculated by the antenna board does not require any additional processing, making the



module ready for use right out of the box.

The ANT-B11

board delivers high-precision direction finding and 2D indoor positioning that make it suitable for commercial tracking solutions and indoor positioning in mid-size spaces. The company also provides applications which allow the user to track an asset and plot their

positions in applications aiming at 2D visualisation.

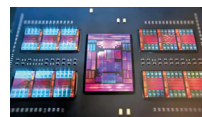
u-Blox

<http://www.u-blox.com>

EMBEDDED

Processors for data centres

The 4th Gen AMD EPYC processors are designed for modern data centres. The processors offer energy efficiency and next-generation architecture which is essential for critical workloads across



cloud, enterprise, and high-performance computing.

The processors

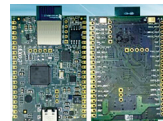
built on high-performing Zen 4 core from AMD also deliver energy efficiency and security. The 4th Gen EPYC processors feature twice the number of encryption keys compared to previous generations, thus keeping the data secure. They can provide up to 2.8 times more performance while consuming up to 54% less power. They also support DDR5 memory and PCIe Gen 5, which offers better speed for AI and ML applications.

AMD

<https://www.amd.com>

ML module for edge devices

The Xeno + WiFi + BLE Nano ML module from Eoxys is developed to meet machine learning (ML) as well as artificial intelligence (AI) application requirements. The module offers



leverage to customers with the onboard MICs and can also attach external MICs using

a separate carrier board. The device is solderable and can be used as a core module for ML edge devices. It offers I2S and I2C connection options for customers to build their battery-operated ML devices. This device can be used for industrial automation, IoT, AI/ML, and deep neural network applications.

Eoxys

<https://www.eoxys.com>